MODULE 23 · DOMAIN 2 - PROCESS

Planning and managing resources

Planning people, equipment and materials from requirements; acquiring them in different organisational structures; and optimising, protecting and releasing them.

DOMAIN II - Process EXAM WEIGHT 41% MODULE 23 of 49

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Plan resources from **requirements**, confirm their **availability**, and manage them as a shared asset.

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Kinds of resource

KIND	EXAMPLES	MAIN CONCERNS
Team resources	Developers, engineers, analysts, contractors	Skills, availability, motivation, development
Equipment	Cranes, test rigs, laptops, licences	Booking, lead time, maintenance, sharing
Materials	Steel, cabling, components	Lead time, storage, waste, quality
Facilities	Offices, labs, test environments	Availability, capacity, conflicts

ON THE EXAM

Physical resources fail projects too: late materials and double-booked environments are common scenario causes.

The resource cycle

01
Plan
How resources will be defined, acquired, managed, released.

02
Estimate
What is needed, how much, and when.

03
Acquire
Obtain the people and physical resources.

04
Lead and manage
Develop the team; control physical use.

05
Release
Return resources on time, with recognition.

ON THE EXAM

Release is part of the cycle. Holding people after their work is done costs the rest of the organisation.

The resource management plan

- **Roles and responsibilities**: role, authority, responsibility, competence, and the RACI from module 07.
- **Acquisition**: how and when people and physical resources are obtained.
- **Development and recognition**: training, team building, rewards.
- **Physical resources**: ordering, storage, inventory, maintenance.
- **Release**: when and how resources return.

> ON THE EXAM
>
> A good resource plan answers who, with what skills, from where, when, and until when.

Estimating resources from requirements

- Start from the **work**: what skills, how many people or units, and exactly when.
- A **resource breakdown structure** groups needs by category and type.
- Include specialist and certified skills that regulation or quality demands.
- Asking for 'three developers' without skills and timing invites the wrong three people, at the wrong time.

> ON THE EXAM
>
> The outline's enabler: define and plan resources based on requirements. Specific needs get the right resources.

Acquiring team resources

METHOD	WHAT IT MEANS
Pre-assignment	Named people are agreed in advance, often in a bid or charter
Negotiation	Agree people and availability with functional managers or other projects
Acquisition	Hire or contract from outside when skills are not available internally
Virtual teams	Use people in other locations, as module 08 covered
Selection criteria	Weigh availability, cost, skills, experience, attitude

ON THE EXAM

In most organisations, people are negotiated for. Build the relationship with functional managers early.

Organisational structure and the PM's authority

STRUCTURE	WHO CONTROLS RESOURCES	PM AUTHORITY
Functional	Functional managers	Little or none; often a coordinator
Weak matrix	Mostly functional managers	Low
Balanced matrix	Shared between PM and functional manager	Moderate
Strong matrix	Mostly the project manager	Moderate to high
Projectised	The project manager	High; team reports to the PM

ON THE EXAM

The weaker the PM's authority, the more resources depend on negotiation and relationships.

Resource calendars and availability

- A **resource calendar** shows when each person or item is actually available.
- Include holidays, part-time allocations, other project commitments, training and leave.
- Confirm availability with the functional manager before building dates on it.
- A schedule that assumes someone full-time, when they are half-time, is wrong from day one.

> ON THE EXAM
>
> The outline's other enabler: manage and optimise resource needs and availability. Start by knowing the availability.

Optimising resources

- Use **resource histograms** to spot over-allocation and idle time.
- **Levelling** and **smoothing**, from module 22, reshape the schedule to fit.
- **Cross-training** reduces dependence on one specialist.
- Share equipment and environments across teams with an agreed booking process.
- Pushing people to 100 percent utilisation creates queues and burnout, not speed.

> ON THE EXAM
>
> Optimise for flow, not for keeping everyone busy. Module 05's motorway applies to people too.

Managing physical resources

- Plan for **lead times**: long-lead items are ordered early, sometimes before design is final.
- **Just-in-time** delivery reduces storage, damage and theft on constrained sites.
- Track inventory, maintenance and calibration of equipment.
- Reduce **waste**: over-ordering and scrap cost money and harm sustainability goals.

> ON THE EXAM
>
> Too early causes storage, damage and waste. Too late stops the critical path. Plan deliveries to the schedule.

When a resource is taken away

- **Assess the impact**: is the person's work on the critical path? For how long?
- **Negotiate** with the functional manager: partial availability, a substitute, different timing.
- Consider options: re-sequence work, cross-train, or bring in a contractor.
- Escalate only if the impact exceeds your authority and negotiation has failed.

> ON THE EXAM
>
> Impact first, then negotiate. Escalating a resource loss before talking to the functional manager usually loses.

Releasing resources well

- Release people **on time**, as planned, and tell their managers early.
- **Recognise** their contribution, and give feedback to their functional managers for appraisals.
- Capture their knowledge before they go, as module 16 described.
- Return equipment and close bookings, licences and access.

> ON THE EXAM
>
> How you release people decides whether functional managers give you good people next time.

Resources on agile teams

- Prefer **stable, dedicated, cross-functional** teams.
- **Bring work to the team**, rather than assembling a new team for each piece of work.
- Plan by team **capacity**, using measured velocity or throughput.
- Moving people between teams to maximise utilisation destroys velocity and trust.

> ON THE EXAM
>
> Agile resourcing optimises the team's flow, not each person's utilisation.

Project Lantern: resources

- **Structure**: a balanced matrix. Clinic staff and IT engineers report to their own managers.
- **Negotiated**: two network engineers at 60 percent for four months, confirmed in writing with the IT manager.
- **Physical**: kiosks delivered just in time to each clinic, because clinics have no storage.
- **Stable**: the app team stays together for the whole project, and beyond.

> ON THE EXAM
>
> Negotiated people, just-in-time equipment, and a stable agile team, in one hybrid project.

Scenario signals and the move

THE SCENARIO SAYS	THE MOVE
A functional manager takes a key person	Assess impact; negotiate options first
Schedule assumed someone full-time	Confirm availability with resource calendars
Which structure gives the PM most authority?	Projectised
Materials damaged or stolen in storage	Just-in-time deliveries matched to the schedule
Offered someone without a required certification	Explain the requirement; negotiate or plan training
People moved between agile teams for utilisation	Stable teams; bring work to the team

ON THE EXAM

Plan from requirements, confirm availability, negotiate, and release well.

> In a balanced matrix organisation, a functional manager tells the project manager that a key engineer will move to another project for three weeks, starting Monday.
>
> The engineer is working on a critical path activity.

What should the project manager do first?

A Assess the impact, then meet the functional manager to negotiate options such as partial availability or a substitute.

B Escalate to the sponsor to stop the move.

C Accept the decision and adjust the schedule.

D Hire a contractor immediately to replace the engineer.

◆ PAUSE AND DECIDE

What should the project manager do first?

A Assess impact; negotiate options.

Understands the cost, then solves it with the person who controls the resource.

B Escalate to the sponsor.

Goes over the functional manager's head before a conversation.

C Accept and adjust.

Gives up without assessing impact or exploring options.

D Hire a contractor now.

Commits cost without authority or analysis. Maybe unnecessary.

THE SIGNAL

A resource taken away: assess impact, then negotiate with the functional manager, before anything else.

> Three weeks into execution, the project manager learns that a security specialist, scheduled full-time on the project, is also committed half-time to another project.
>
> Her work is already falling behind.

What should the project manager have done to prevent this?

A Crashed her activities at the start of the project.

B Confirmed her availability with her functional manager and a resource calendar when planning.

C Added more contingency reserve to the budget.

D Included her in the RACI chart.

◆ PAUSE AND DECIDE

What should the project manager have done to prevent this?

A Crash from the start.

Crashing is a response to lateness, not a way to discover availability.

B Confirm availability with a resource calendar.

Plans on real availability, so the schedule is right from day one.

C More contingency.

Money does not create a person's missing half-time.

D Put her in the RACI.

A RACI records responsibility, not availability.

THE SIGNAL

A schedule built on assumed availability is wrong from the start. Confirm it with a resource calendar.

> A project manager is choosing between two job offers. In one, team members report directly to the project manager, who controls the budget and assignments.
>
> In the other, team members report to department heads, who decide their priorities.

The first role describes which organisational structure?

A Projectised

B Weak matrix

C Functional

D Balanced matrix

◆ PAUSE AND DECIDE

The first role describes which organisational structure?

A Projectised

Team reports to the PM, who controls budget and assignments: high authority.

B Weak matrix

Functional managers keep most control. The PM coordinates.

C Functional

This describes the second role, where department heads decide priorities.

D Balanced matrix

Authority is shared, not held by the PM alone.

THE SIGNAL

Team reports to the PM, PM controls budget: projectised. Department heads decide: functional or weak matrix.

> On a city-centre building site with almost no storage space, all the cladding panels were delivered in the first week. Several have been damaged by weather and site traffic, and two pallets were stolen.
>
> The cladding will be installed over the next four months.

What should the project manager do for the remaining materials?

A Rent a larger secure storage area off site for all future deliveries.

B Arrange just-in-time deliveries, scheduled to match the installation sequence.

C Increase the insurance cover and continue as planned.

D Order extra panels to cover future losses.

◆ PAUSE AND DECIDE

What should the project manager do for the remaining materials?

A Rent off-site storage.

Adds cost and handling, and treats the symptom rather than the delivery pattern.

B Just-in-time deliveries to the sequence.

Removes the storage problem at its source, on a constrained site.

C More insurance.

Pays for losses; does not prevent them or the delays they cause.

D Order extra panels.

Plans for waste, adds cost, and harms sustainability goals.

THE SIGNAL

No storage and materials needed over months: deliver just in time to the installation schedule.

> The resource plan for a pipeline project requires welders certified to a specific safety standard, as regulation demands.
>
> The functional manager offers an experienced welder who does not hold that certification, saying 'he is just as good'.

What should the project manager do?

A Accept the welder, since he is experienced.

B Hire a certified welder externally without telling the functional manager.

C Explain the regulatory requirement, and negotiate for a certified welder or a plan to certify this one in time.

D Remove the certification requirement from the resource plan.

◆ PAUSE AND DECIDE

What should the project manager do?

A Accept him.

Breaches a regulatory requirement, whatever his skill.

B Hire externally, secretly.

Bypasses the functional manager and the agreed acquisition route.

C Explain; negotiate a certified welder or certification.

Keeps compliance, works with the manager, and may develop the person too.

D Drop the requirement.

The requirement comes from regulation. It is not the PM's to remove.

THE SIGNAL

A regulatory skill requirement is not negotiable. The route to meeting it is.

> A department head proposes moving developers between four agile teams each week, to wherever demand is highest, so that nobody is ever idle.
>
> The teams' velocities have been stable for months.

What should the project manager recommend?

A Adopt the proposal, since higher utilisation should increase output.

B Assign every developer half-time to two teams, to balance the load.

C Refuse the proposal, since agile teams must never change.

D Explain the value of stable teams, and propose bringing work to the teams based on their capacity.

◆ PAUSE AND DECIDE

What should the project manager recommend?

A Adopt it.

Every move resets team dynamics and knowledge; output falls, not rises.

B Half-time on two teams.

Doubles the task switching and halves each team's stability.

C Refuse outright.

Too absolute, and ignores the manager's real concern about demand.

D Stable teams; bring work to them.

Meets the concern about demand without destroying team performance.

THE SIGNAL

Maximising each person's utilisation damages the team's flow. Move work, not people.

Plan from requirements. Confirm availability. **Negotiate**, then release well.

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